

Retina Image Classification and Segmentation Pipeline Documentation

1. Objective

To build a robust deep learning pipeline for:

- **Classification:** Predicting Diabetic Retinopathy grade (5 classes) and Macular Edema risk (3 classes).
 - **Segmentation:** Identifying 5 anatomical/pathological structures in fundus images.
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2. Data Preprocessing

2.1 Resizing

- All input images and masks resized to **512x512** to:
 - Maintain uniformity
 - Fit GPU memory
 - Preserve sufficient anatomical detail

2.2 Augmentation for Training Data

Performed using **Albumentations**:

```
A.Compose([
    A.HorizontalFlip(p=0.5),
    A.VerticalFlip(p=0.5),
    A.Rotate(limit=20, p=0.5),
    A.RandomBrightnessContrast(p=0.3),
    A.Resize(512, 512),
])
```

- Avoided extreme color distortions.
- Increased dataset diversity.

2.3 Saving Replicas

- Augmented images were saved as new copies to address dataset imbalance and scarcity.

2.4 Normalization

- Normalized to ImageNet mean/std during training and inference (only during model input)

2.5 Test Data Processing

- No augmentation applied
 - Only resized and normalized before inference
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3. Models Used

3.1 Image Classification Model

Backbone: EfficientNetB7 (pretrained)

- Justification:
 - State-of-the-art performance on medical image tasks
 - Efficient use of parameters
 - Strong feature extractor

Architecture:

- Two classification heads:
 - **Retinopathy** (5-class softmax)
 - **Edema** (3-class softmax)

3.2 Image Segmentation Model

Model: U-Net

- Input: RGB Image
- Output: Multi-class segmentation mask (6 classes incl. background)

Justification:

- U-Net is the gold standard for biomedical segmentation
 - Works well on small datasets with limited annotations
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4. Overfitting Prevention Techniques

- **L2 Regularization:** Added to optimizer
 - **Dropout:** Used in classifier FC layers
 - **Early Stopping:** Stopped training when validation loss plateaued
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5. Training Summary

Image Classification

Achieved high accuracy on training & validation:

- **Final Val Acc:** ~99% (Retino), ~99.7% (Edema)
- **Used Augmentation** + EfficientNetB7 + Dropout + Early stopping

Image Segmentation

- **Final Val Loss:** ~0.03
 - Model generalized well (IoU ~0.47 on test set)
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6. Evaluation Results

Classification

Test Loss: 3.8181

- **Test Accuracy:** Retino - 63.11%, Edema - 77.67%

Classification Report:

Retinopathy

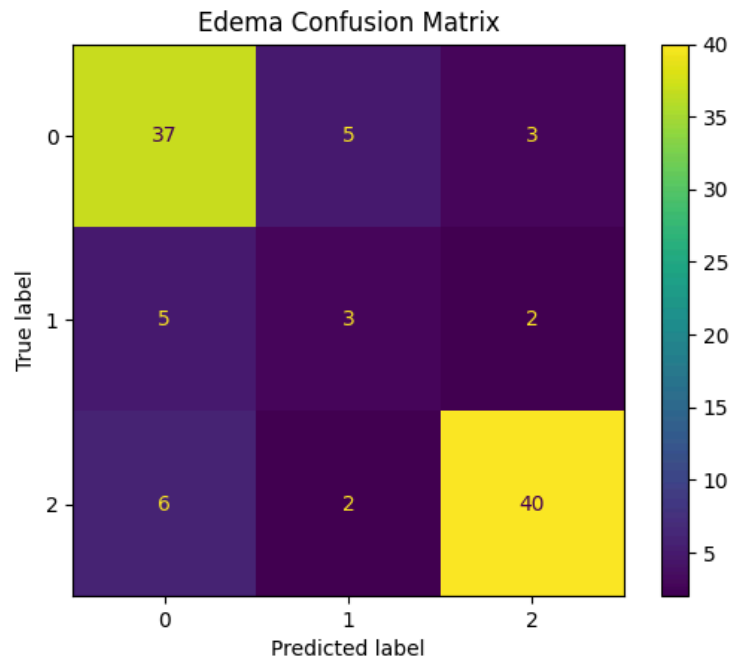
	precision	recall	f1-score	support
0	0.71	0.85	0.77	34
1	0.20	0.20	0.20	5
2	0.67	0.56	0.61	32
3	0.52	0.58	0.55	19
4	0.67	0.46	0.55	13
accuracy			0.63	103
macro avg	0.55	0.53	0.54	103
weighted avg	0.63	0.63	0.62	103

Edema

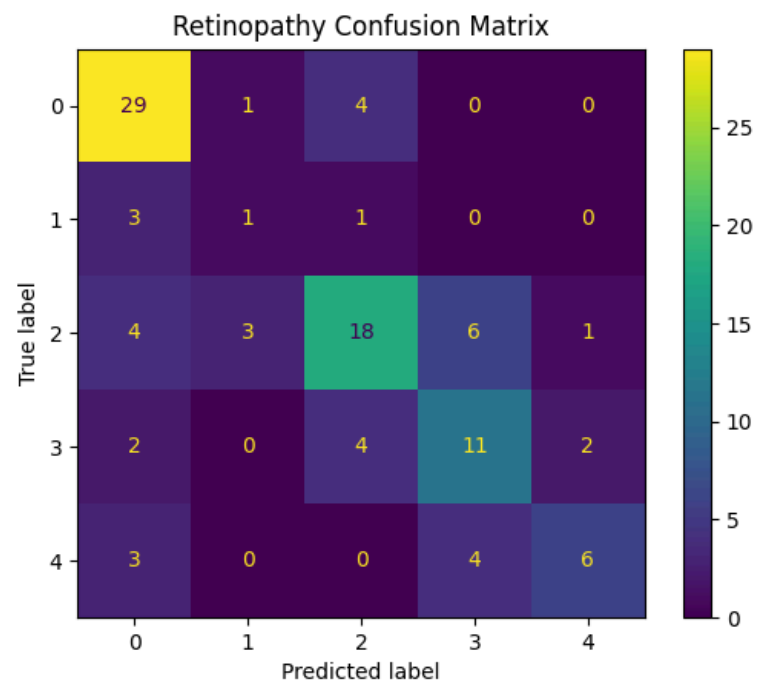
	precision	recall	f1-score	support
0	0.77	0.82	0.80	45
1	0.30	0.30	0.30	10
2	0.89	0.83	0.86	48
accuracy			0.78	103
macro avg	0.65	0.65	0.65	103
weighted avg	0.78	0.78	0.78	103

Confusion Matrices:

- Edema:



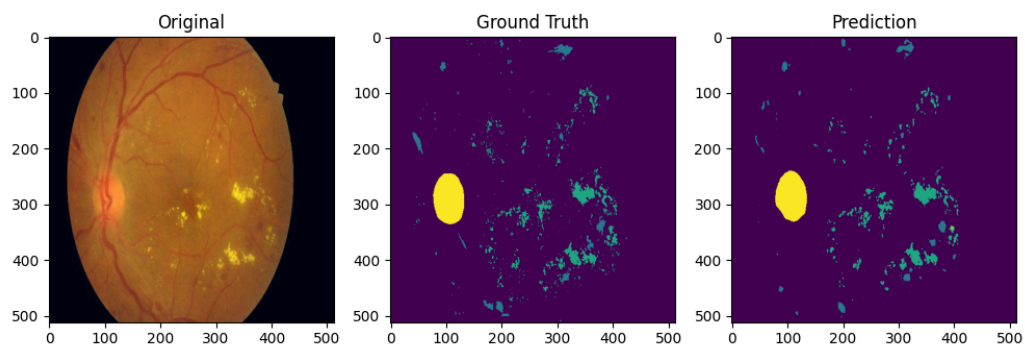
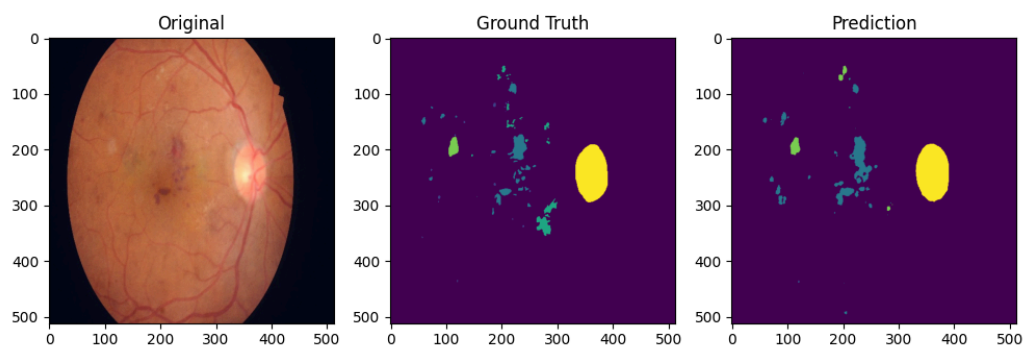
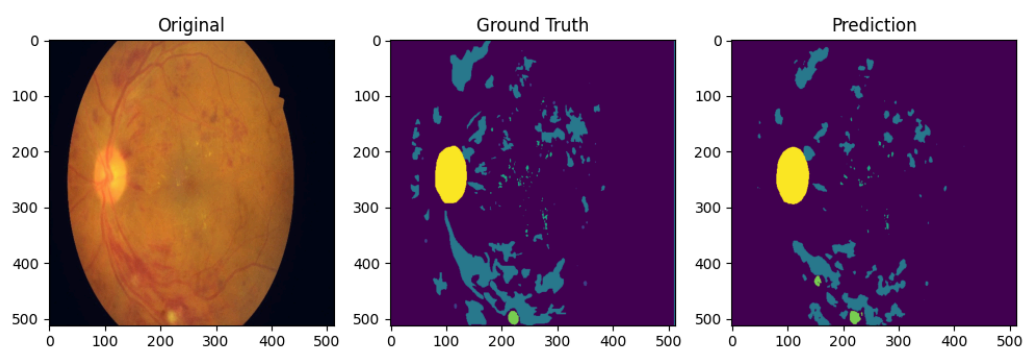
- Retinopathy:



Segmentation

- Test Avg IoU: 0.4783

Example Output:



7. Inference Script:

Functionality

- Accepts an image path using:

```
python retina_predictor.py -i path/to/image.jpg
```

Steps

1. Loads classification and segmentation models
2. Runs both models in **parallel threads**
3. Displays:
 - **Predicted Labels** for Retino & Edema
 - **Color-coded Segmentation Mask** overlaid on original image with legend

Output Example:

Retinopathy Grade: Moderate

Macular Edema Risk: Severe

Segmentation display includes anatomical region color codes:

- Microaneurysms: Red
- Haemorrhages: Green
- Hard Exudates: Yellow
- Soft Exudates: Magenta
- Optic Disc: Cyan

8. Directory Structure

```
Retina_ImageClassification_and_Segmentation/  
|  
|-- retina_image_classification_training.py  
|-- retina_image_classification_evaluation.py  
|-- retina_image_segmentation_training.py  
|-- retina_image_segmentation_evaluation.py  
|-- retina_preprocessing.py  
|-- retina_predictor.py  
|-- retina_models.py  
|-- processed_retina/  
|   |-- classification/  
|   |-- segmentation/  
|-- checkpoints/  
|-- logs/  
|-- visuals/
```
